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Inventor(s)	Ge; Xinyang et al.

Protecting commercial off-the-shelf program binaries from piracy using hardware enclaves

Abstract

This disclosure describes systems and methods for protecting commercial off-the-shelf software program code from piracy. A software program may include an executable file. The executable file may include code and data. A platform may modify the executable file such that the data may be placed at a location in memory not based on a fixed distance from the code. The platform may modify the executable file to indicate that the code should be loaded in a hardware enclave and at least a portion of the data should be loaded in the memory outside the hardware enclave. The platform may encrypt the code and provide it to a computing device.

Inventors: Ge; Xinyang (Redmond, WA), Cui; Weidong (Redmond, WA), Niu; Ben (Bellevue, WA), Chen; Ling Tony (Bellevue, WA)

Applicant: Microsoft Technology Licensing, LLC (Redmond, WA)

Family ID: 73793819

Assignee: Microsoft Technology Licensing, LLC (Redmond, WA)

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Primary Examiner: Wright; Bryan F

Attorney, Agent or Firm: Ray Quinney & Nebeker P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/970,185, filed Oct. 20, 2022, which is a continuation of U.S. patent application Ser. No. 16/734,197, filed Jan. 3, 2020, now U.S. Pat. No. 11,500,969, which are hereby incorporated by reference in their entireties.

BACKGROUND

(1) Computing devices may be used to perform a wide variety of tasks. Software developers may design and create software programs for use on computing devices. Users of computing devices may use software applications to perform tasks. For example, a user of a computing device may use an internet browser application to access information on the internet and then use a presentation application to create a presentation communicating information obtained from the internet. Software applications may include a number of different files, including executable files.

(2) Software developers may worry about software piracy. Software piracy may occur when an end user purchases or obtains a copy of an application, makes unauthorized copies of the application, and uses the unauthorized copies or distributes them to third parties. Stopping piracy may be difficult because the attacker often has full control of the computing device and system that runs the application.

SUMMARY

(3) In accordance with one aspect of the present disclosure, a method is disclosed for protecting software from piracy. The method includes receiving a binary file comprising code and data. The code includes a data reference that is based on a fixed distance in memory between the code and the data. The method further includes modifying the data reference in the code. The method further includes modifying the binary file to indicate that the code of the binary file should be loaded in a hardware enclave and at least a portion of the data of the binary file should be loaded in the memory outside the hardware enclave.

(4) The method may further include encrypting the code of the binary file but not the data of the binary file. The method may further include receiving, from a computing device, a request for a decryption key. The computing device may include the hardware enclave. The method may further include providing, to the computing device, the decryption key.

(5) The hardware enclave may include a defined region of memory of the computing device that provides protection from instructions not residing in the hardware enclave.

(6) The method may further include authenticating, before providing the decryption key, a

processor signature. A processor of the computing device may have generated the processor signature and the request for the decryption key may include the processor signature. The method may further include validating, before providing the decryption key, a hash. The request for the decryption key may include the hash.

(7) The method may further include identifying the data reference in the code of the binary file without access to source code or debug symbols for the binary file. Identifying the data reference may include accessing relocation information about the data reference.

(8) In accordance with another aspect of the present disclosure, a system is disclosed for facilitating protecting a software program from piracy. The system includes one or more processors, memory in electronic communication with the one or more processors, and instructions stored in the memory. The instructions are executable by the one or more processors to receive a binary file comprising code and data. The code includes a data reference that is based on a fixed distance in device memory between the code and the data. The instructions are also executable by the one or more processors to modify the data reference in the code. The instructions are also executable by the one or more processors to modify the binary file to indicate that the code of the binary file should be loaded in a hardware enclave and at least a portion of the data of the binary file should be loaded in the device memory outside the hardware enclave.

(9) Modifying the binary file to indicate that the code of the binary file should be loaded in the hardware enclave and at least a portion of the data of the binary file should be loaded in the device memory outside the hardware enclave includes modifying the binary file to include a separation header.

(10) The instructions may be further executable by the one or more processors to encrypt the code of the binary file but not the data of the binary file. The instructions may be further executable by the one or more processors to receive, from a computing device, a request for a decryption key. The computing device may include the hardware enclave. The instructions may be further executable by the one or more processors to provide, to the computing device, the decryption key. The hardware enclave may include instructions to mark the hardware enclave as non-readable before executing the code.

(11) The instructions may be further executable by the one or more processors to authenticate, before providing the decryption key, a processor signature. A processor of the computing device may have generated the processor signature and the request for the decryption key may include the processor signature. The instructions may be further executable by the one or more processors to validate, before providing the decryption key, a hash. The request for the decryption key may include the hash and validating the hash may include comparing the hash to a verified hash value.

(12) The instructions may be further executable by the one or more processors to identify the data reference in the code of the binary file without access to source code or debug symbols for the binary file. Identifying the data reference comprises accessing relocation information about the data reference.

(13) In accordance with another aspect of the present disclosure, a computer-readable medium is disclosed. The computer-readable medium includes instructions that are executable by one or more processors to cause a computing system to receive a binary file that includes code and data. The code includes a data reference that is based on a fixed distance in memory between the code and the data. The instructions are also executable by the one or more processors to modify the data reference in the code. The instructions are also executable by the one or more processors to modify the binary file to indicate that the code of the binary file should be loaded in a hardware enclave and at least a portion of the data of the binary file should be loaded in the memory outside the hardware enclave.

(14) Modifying the binary file to indicate that the code of the binary file should be loaded in the hardware enclave and at least a portion of the data of the binary file should be loaded in the memory outside the hardware enclave may include modifying the binary file to include a separation

header.

(15) The computer-readable medium may further include additional instructions that are executable by the one or more processors to cause the computing system to encrypt the code of the binary file but not the data of the binary file. The additional instructions may also be executable by the one or more processors to cause the computing system to receive, from a computing device, a request for a decryption key. The computing device may include the hardware enclave. The additional instructions may also be executable by the one or more processors to cause the computing system to provide, to the computing device, the decryption key. The hardware enclave may include instructions to mark the hardware enclave as non-readable before executing the code.

(16) The computer-readable medium may further include additional instructions that are executable by the one or more processors to cause the computing system to measure a hash of the startup code and the executable code.

(17) The computer-readable medium may further include additional instructions that are executable by the one or more processors to cause the computing system to authenticate, before providing the decryption key, a processor signature. A processor of the computing device may have generated the processor signature. The request for the decryption key may include the processor signature. The additional instructions that are executable by the one or more processors may also cause the computing system to validate, before providing the decryption key, a hash. The request for the decryption key may include the hash. Validating the hash may include comparing the hash to a verified hash value.

(18) Modifying the data reference may not require access to source code or debug symbols for the binary file.

(19) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(20) Additional features and advantages will be set forth in the description that follows. Features and advantages of the disclosure may be realized and obtained by means of the systems and methods that are particularly pointed out in the appended claims. Features of the present disclosure will become more fully apparent from the following description and appended claims, or may be learned by the practice of the disclosed subject matter as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the manner in which the above-recited and other features of the disclosure can be obtained, a more particular description will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. For better understanding, the like elements have been designated by like reference numbers throughout the various accompanying figures. Understanding that the drawings depict some example embodiments, the embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1 illustrates an example system for utilizing a hardware enclave to protect a software program from software piracy.

(3) FIGS. 2A and 2B illustrate an example file that may be modified in a way to facilitate protecting the example file from software piracy.

(4) FIG. 3 illustrates an example method for modifying a file to facilitate protecting the file from software piracy.

(5) FIG. 4 illustrates an example method for loading a file into memory to facilitate protecting the

file from software piracy.

(6) FIG. 5 illustrates an example method for authenticating information and providing a decryption key to facilitate protecting a file from software piracy.

(7) FIG. 6 illustrates an example method for initiating execution of a program to facilitate protecting the program from software piracy.

(8) FIG. 7 illustrates certain components that may be included in a computing system.

DETAILED DESCRIPTION

(9) A software developer may create an application for use on computing devices. The application may include sets of instructions that cause computing devices to perform certain tasks. The software developer may invest significant time and money into designing and creating the application's code. The software developer may intend to profit from sales (by the software developer or a distributor) of authorized copies of the application. As a result, the software developer may want to protect the application from piracy. Software piracy may occur when a user purchases or obtains a copy of the application, makes unauthorized copies of the application, and uses the unauthorized copies or distributes them to third parties. The third parties may use the unauthorized copies of the application instead of purchasing an authorized copy from the software developer or the software developer's distributor. Thus, software piracy may directly impact the developer's revenues and profits. Stopping piracy, however, may be difficult because the attacker often has full control of the system that runs the application.

(10) This disclosure describes systems and methods for utilizing hardware enclaves to protect program code contained in commercial off-the-shelf (COTS) programs from software piracy. A hardware enclave may be a defined region of memory whose contents cannot be read or saved by any software, including the operating system and even hypervisor, not stored in the hardware enclave. A set of instruction codes, such as Intel® Software Guard Extensions (Intel® SGX) or AMD Secure Encrypted Virtualization (SEV), built into a processor may support creation and use of the hardware enclave. The set of instructions and the hardware enclave may allow a remote party to run an unmodified program with confidentiality and/or integrity protection even if the rest of the system is malicious. Code and data stored in the hardware enclave may trust only the hardware enclave and may treat any process outside the hardware enclave, including an operating system or a hypervisor, as potentially hostile. The processor may encrypt information stored in the hardware enclave and decrypt the information on-the-fly within the processor itself.

(11) Although hardware enclaves may provide opportunities to protect information from malicious systems (requiring trust only in the processor), a software developer may not utilize hardware enclaves because doing so requires that the software developer modify its programs. But the described systems and methods require little to no help from the developers of a program and do not require access to source code or rely on recompiling a program. Instead, the described system and methods may be applied directly to commercial off-the-shelf (COTS) program binaries by a person or entity other than the software developer. Thus, aspects of the described systems and methods may be provided as a service to software developers by a software distribution platform, an operating system developer, or other trusted parties. As a result, the described systems and methods may improve utilization of hardware enclaves to defeat piracy for both new and legacy software. Of course, a software developer may also perform aspects of the described systems and methods rather than rely on a third party.

(12) Assume a software developer provides a COTS application to a platform to distribute and sell the application to consumers. The application may include one or more binaries that include executable code. In the case of a PC application, the application may include one or more EXE files and one or more DLL files. Each binary may include one or more code sections (sections of executable instructions) and one or more data sections containing information that the executable instructions reference and use. To run properly the binaries may require that the data sections be placed in memory at a predetermined and fixed distance from the code sections. The application or

the binaries may define the predetermined and fixed distance.

(13) Before making the application available to purchasers, the platform may modify each binary that includes executable code. The platform may modify each binary in up to three ways: (1) separate the code sections from the data sections; (2) encrypt the code sections, and (3) add a header.

(14) First, for each binary file that includes executable code, the platform may separate the code sections from the data sections such that the code and the data can be loaded at two locations in memory that are an arbitrary distance apart (as opposed to the predetermined distance required before separation). Separating the code from the data may involve modifying all data references in the code that assume a fixed distance between the code and the data. The platform may separate the code from the data without relying on any information not included in the binaries. Thus, the platform may separate the code from the data without accessing source code or debug symbols of the binaries. Separating the code from the data may facilitate loading all code sections from multiple binaries adjacently into a single, continuous memory region.

(15) Second, after the platform separates the code from the data, the platform may encrypt the code sections contained in each modified binary. The platform may leave the data sections of the binaries as unencrypted plain text.

(16) Third, the platform may add a header or other information to one or more of the modified binaries. The header may communicate to an operating system that the code sections in the modified binaries have been separated from the data sections in the modified binaries. The header may also communicate that the code sections should be loaded into a hardware enclave and that the data sections should be loaded outside the hardware enclave.

(17) After modifying the binaries, the platform may sell or distribute the application (with the modified binaries) to the public. A user may download the application to disk memory on a computing device that also includes a hardware enclave. When the user launches the application, an operating system of the computing device may load the binaries of the application into memory. The operating system may be designed to recognize when the code sections in the binaries are separate from the data sections. The operating system may also be designed to recognize when the code sections should be loaded into the hardware enclave. The operating system may determine, based on the header, that the code and data sections are separate and that the code sections should be loaded into the hardware enclave. After loading all the code sections from the modified binaries into the hardware enclave in one continuous range, the operating system may load all the data sections from the modified binaries into process memory outside of the hardware enclave.

(18) To load the modified binaries in this way the operating system may include a loader that is different from a standard loader. A loader may be a part of the operating system designed to move executable files for an application from disk storage to process memory and to bootstrap the startup process for the application. A standard loader may load the code sections and the data sections into memory together and may not recognize information in the modified binaries indicating that the code sections should be loaded together. A loader included in the operating system described above, however, is designed to load the code sections of the modified binaries together in the hardware enclave and the data sections outside the hardware enclave. But even though the loader of the operating system described above may be different from a standard loader, the operating system kernel is not customized or changed. Therefore, the described systems and methods may require only minimal modifications to existing operating systems.

(19) In addition to loading the modified binaries, the operating system may load startup code into the hardware enclave. The startup code may be included in the operating system and may not be encrypted. In the alternative, the platform may provide the startup code to the computing device with the application.

(20) Once the loader has loaded everything, the operating system may transfer control to the startup code. The startup code may cause the processor to measure a hash of the code residing in the

hardware enclave and sign the hash with a processor signature. The processor may then establish a secure communication channel with an authentication server of the platform and send the signed hash to the authentication server. The authentication server may verify the processor signature to ensure the processor is authentic. The authentication server may also verify the hash. If the authentication server verifies both the signature and the hash, the authentication server may send a key for decrypting the code to the computing device. The startup code may use the key to decrypt the code residing in the hardware enclave.

(21) Before the application runs, the startup code may mark all hardware enclave memory as non-readable to code stored in the hardware enclave in order to protect the application from leaking code content. Otherwise, an attacker may try to trick an application to leak code content by manipulating data pointers that are not protected by the hardware enclave. The data sections stored outside the hardware enclave may contain pointers. An attacker may be able to change the pointers in the data sections to point to the code sections. If the code does not check whether the data is pointing to the code, the code may leak its contents if the hardware enclave were not marked as non-readable.

(22) Notwithstanding the potential to combat software piracy, hardware enclaves have not been widely adopted. One reason, as noted above, may be that utilizing hardware enclaves requires software developers to modify existing software. Typically, a software developer must modify a program to make it aware of the hardware enclave to leverage capabilities of the hardware enclave. But the disclosed systems and methods do not require the software developer to modify a program. Nor do the disclosed systems and methods require that the software developer provide source code. Instead, the disclosed systems and methods allow a platform or service provider to modify the program to utilize the hardware enclave without requiring help or additional information from the software developer. In this way, the described systems and methods allow a software distribution platform to offer an end-to-end anti-piracy solution for third-party applications without relying on the software developers.

(23) Another reason hardware enclaves may not have been widely adopted is a lack of support from major operating systems. But the described systems and methods preserve the compatibility with the commodity operating systems because they do not affect how operating systems access the program's data to service certain system calls (e.g., loading file content to a program's memory). The program's data still resides in the normal process memory, which is both readable and writable. Thus, the disclosed systems and methods do not require changes to today's operating system kernel. Instead, an operating system developer need only modify the operating system loader, making this approach amenable for real-world adoption on multiple platforms (e.g., Windows, Mac, etc.).

(24) Another reason hardware enclaves may not have been widely adopted is the limited amount of enclave memory. The described systems and methods mitigate the memory pressure by using enclave memory for only a program's code sections, which are typically significantly smaller than a program's data sections. Moreover, software pirates may be more interested in application code rather than data.

(25) FIG. 1 illustrates an example system **100** in which the techniques disclosed herein for protecting software from piracy may be utilized. The system **100** may include a platform **102**, an authentication server **146**, and a computing device **104**. The platform **102** and the computing device **104** may be connected through a network **142**. The computing device **104** and the authentication server **146** may also be connected through the network **142**.

(26) A software developer **106** may utilize the system **100**. The software developer **106** may create a program **108a**. The program **108a** may be a set of instructions and information designed to perform one or more functions or tasks. The program **108a** may be a commercial-off-the-shelf (COTS) software program that can be run on a computing device, such as the computing device **104**.

(27) The program **108a** may include one or more files, such as file **110a** and file **112a**. The program

108a may include additional files not shown in FIG. 1. In other designs, the program **108a** may include only one file. The files **110a**, **112a** may be unencrypted binary files. The files **110a**, **112a** may include executable code and data. Executable code may be instructions that a processor can understand and execute. The executable code in the files **110a**, **112a** may include references to the data included in the files **110a**, **112a**. The files **110a**, **112a** may be designed such that to run on a computing device, the data included in the files **110a**, **112a** must be stored at a fixed and predetermined distance in memory from the code included in the files **110a**, **112a**. The program **108a** or the files **110a**, **112a** may define the fixed and predetermined distance.

(28) The platform **102** may provide software programs for distribution and sale. The platform **102** may include an online store from which users can download software programs. The platform **102** may receive the program **108a** from the software developer **106**. The software developer **106** may provide the program **108a** to the platform **102** for the purpose of having the platform **102** sell authorized copies of the program **108a** to users of computing devices, such as the computing device **104**. The software developer **106** may worry that someone may purchase the program **108a** through the platform **102**, make unauthorized copies of the program **108a**, and then distribute the unauthorized copies to third parties without compensating the software developer **106**. The platform **102** may provide a service to the software developer **106** to facilitate protecting the program **108a** from software pirates.

(29) The platform **102** may include a program **108b**. The program **108b** may be a modified version of the program **108a**. The program **108b** may include a modified file **110b** and a modified file **112b**. The modified file **110b** may be a modified version of the file **110a**, and the modified file **112b** may be a modified version of the file **112a**. The modified files **110b**, **112b** may be binary files. The platform **102** may have modified the program **108a** to create the program **108b** in order to facilitate protecting the program **108a** from software piracy. In the alternative, the software developer **106** or an entity other than the platform **102** may modify the program **108a** to create the program **108b**.

(30) The modified file **110b** may include encrypted code **114b-1**, encrypted code **114b-2**, data **116b-1**, data **116b-2**, and a header **118b-1**. The encrypted code **114b-1** and the encrypted code **114b-2** may be modified versions of the code included in the file **110a**. The encrypted code **114b-1**, **114b-2** may allow a computing device to execute the modified file **110b** when the data **116b-1**, **116b-2** are located at an arbitrary distance in memory from the encrypted code **114b-1**, **114b-2**. Furthermore, the encrypted code **114b-1**, **114b-2** may be encrypted such that a key is required to decrypt and run the encrypted code **114b-1**, **114b-2**. Unlike the encrypted code **114b-1**, **114b-2**, the data **116b-1**, **116b-2** may be unencrypted.

(31) The header **118b-1** may include information about the modified file **110b**. The header **118b-1** may indicate that the modified file **110b** includes code and data that have been separated. In other words, the header **118b-1** may communicate that the data **116b-1**, **116b-2** may be placed in memory at an arbitrary distance from the encrypted code **114b-1**, **114b-2**. The header **118b-1** may indicate that the encrypted code **114b-1**, **114b-2** should be placed in a hardware enclave and the data **116b-1**, **116b-2** should be placed in memory outside a hardware enclave. Although the header **118b-1** is shown in the modified file **110b**, the information contained in the header **118b-1** may instead be included in the program **108b** outside the modified file **110b**.

(32) The modified file **112b** may include encrypted code **114b-3**, data **116b-3**, and a header **118b-2**. The encrypted code **114b-2** may be a modified version of the code included in the file **112a**. The encrypted code **114b-3** may allow a computing device to execute the modified file **112b** when the data **116b-3** is stored an arbitrary distance in memory from the encrypted code **114b-3**. Furthermore, the encrypted code **114b-3** may be encrypted. Unlike the encrypted code **114b-3**, the data **116b-3** may be unencrypted.

(33) The header **118b-2** may include information about the modified file **112b**. The header **118b-2** may indicate that the modified file **112b** includes code and data that have been separated. In other words, the header **118b-2** may communicate that the data **116b-3** may be placed in memory at an

arbitrary distance from the encrypted code **114b-3**. The header **118b-2** may indicate that the encrypted code **114b-3** should be placed in a hardware enclave and the data **116b-3** should be placed in memory outside a hardware enclave. Although the header **118b-2** is shown in the modified file **112b**, the information contained in the header may instead be included in the program **108b** outside the modified file **112b**.

(34) The computing device **104** may download software programs from the platform **102** for use on the computing device **104**. A user of the computing device **104** may use downloaded software programs to perform functions and tasks. The computing device **104** may include a program **108c** stored on disk storage **140**. The program **108c** may be a copy of the program **108b**. The computing device **104** may have downloaded the program **108c** from the platform **102** through the network **142**.

(35) The program **108c** may include a modified file **110c** and a modified file **112c**. The modified file **110c** may be a copy of the modified file **110b**, and the modified file **112c** may be a copy of the modified file **112b**. The modified file **110c** may include encrypted code **114c-1** (which may be a copy of the encrypted code **114b-1**), encrypted code **114c-2** (which may be a copy of the encrypted code **114b-2**), data **116c-1** (which may be a copy of the data **116b-1**), data **116c-2** (which may be a copy of the data **116b-2**), and a header **118c-1** (which may be a copy of the header **118b-1**). The modified file **112c** may include encrypted code **114c-3** (which may be a copy of the encrypted code **114b-3**), data **116c-3** (which may be a copy of the data **116b-3**), and a header **118c-2** (which may be a copy of the header **118b-2**).

(36) A user of the computing device **104** may cause the computing device **104** to initiate the program **108c**. An operating system **126** of the computing device **104** may use a loader **128** to load the program **108c** into memory **130** of the computing device. The operating system **126** may be a program that manages hardware and software on a computing device, such as the computing device **104**. The platform **102** may have developed the operating system **126**. The loader **128** may be designed to determine whether the modified files **110c**, **112c** include code that has been separated from data. The loader **128** may determine from the headers **118c-1**, **118c-2** that the modified files **110c**, **112c** include code that has been separated from data.

(37) The loader **128** may also determine from the headers **118c-1**, **118c-2** that the code included in the modified files **110c**, **112c** should be loaded into a hardware enclave **132** of the memory **130**. The hardware enclave **132** may be a defined portion of the memory **130** that has confidentiality and integrity protection from instructions not residing in the hardware enclave **132**. A processor **136** of the computing device **104** may manage, protect, and support the hardware enclave **132**.

(38) The loader **128** may load the encrypted code **114c-1**, **114c-2**, **114c-3** into the hardware enclave **132**. The loader **128** may load the encrypted code **114c-1**, **114c-2**, **114c-3** in a continuous range of the hardware enclave **132**. The loader **128** may also load startup code **134** into the hardware enclave **132**. The startup code **134** may be included in the operating system **126**. In the alternative, the computing device **104** may obtain the startup code **134** from the platform **102** or as part of the program **108b**. The startup code **134** may be unencrypted. The loader **128** may load the data **116c-1**, **116c-2**, **116c-3** into the memory **130** outside of the hardware enclave **132**. The data **116c-1**, **116c-2**, **116c-3** may be located an arbitrary distance in the memory **130** from the encrypted code **114c-1**, **114c-2**, **114c-3**.

(39) Once the loader **128** has loaded the modified files **110c**, **112c** into the memory **130**, the operating system **126** may initiate the startup code **134**. The startup code **134** may include instructions to cause the processor to perform remote attestation in order to obtain a key to decrypt the encrypted code **114c-1**, **114c-2**, **114c-3**, such as a decryption key **124**. As part of performing remote attestation, the startup code **134** may include instructions to cause the processor **136** to provide information attesting to the authenticity of the contents of the hardware enclave **132**. For example, the startup code **134** may include instructions to cause the processor **136** of the computing device **104** to measure a hash of the encrypted code **114c-1**, **114c-2**, **114c-3**. The

processor **136** may also include the startup code **134** in the hash measurement. The processor may use a public hash function to measure the hash of the encrypted code **114c-1**, **114c-2**, **114c-3** and the startup code **134**.

(40) As part of performing remote attestation, the processor **136** may provide information attesting to its authenticity. For example, the processor **136** may include a signature module **138**, and the processor **136** may cause the signature module **138** to sign the hash with a processor signature. The signature module **138** may sign the hash using a private certificate maintained by the processor **136**.

(41) As part of performing remote attestation, the startup code **134** may cause the processor **136** to establish a communication channel with the authentication server **146** over the network **142**. The communication channel may be a secure communication channel. The processor **136** may send information to the authentication server **146** attesting to the authenticity of the processor **136** and the contents of the hardware enclave **132**. For example, the processor **136** may send the signed hash to the authentication server **146** over the communication channel. The processor **136** may send the signed hash to the authentication server **146** as part of a request for the decryption key **124**. The decryption key **124** may enable decryption of the encrypted code **114b-1**, **114b-2**, **114b-3**. In some designs, the authentication server **146** may be included in the platform **102**. In other designs, the authentication server **146** may be separate from the platform **102**. In that case, the authentication server **146** may have received the decryption key **124** from the platform **102**.

(42) The authentication server **146** may verify the processor signature received from the processor **136**. The authentication server **146** may use a public certificate **148** to verify the processor signature. The authentication server **146** may verify the processor signature to determine that the processor **136** is authentic and can be trusted. If the authentication server **146** cannot verify the processor signature, the processor **136** may be malicious or controlled by an attacker. In that case, the processor **136** may not have placed the encrypted code **114b-1**, **114b-2**, **114b-3** in a secure hardware enclave. When the authentication server **146** cannot verify the processor signature, the authentication server **146** may not provide the decryption key **124** to the computing device **104**. Otherwise, a user of the computing device **104** may be able to cause the computing device **104** to place the encrypted code **114b-1**, **114b-2**, **114b-3** in a non-secure portion of the memory **130**, use the decryption key **124** to decrypt the encrypted code **114b-1**, **114b-2**, **114b-3**, and make unauthorized copies of the decrypted code.

(43) The authentication server **146** may verify the hash. The authentication server **146** may include a verified hash value **150**. The authentication server **146** may determine whether the hash received from the processor **136** matches the verified hash value **150**. The platform **102** may have determined the verified hash value **150** when the platform **102** generated the modified files **110b**, **112b**. The platform **102** may have had access to the startup code **134** to generate the verified hash value **150**. The authentication server **146** may verify the hash to determine that the computing device **104** has not modified the encrypted code **114b-1**, **114b-2**, **114b-3** or the startup code **134**. If the authentication server **146** cannot verify the hash, the authentication server **146** may not provide the decryption key **124** to the computing device. Otherwise, a user of the computing device **104** may be able to modify the startup code **134** or the encrypted code **114b-1**, **114b-2**, **114b-3** to include instructions to provide information about the decrypted contents of the hardware enclave **132**. By verifying the hash, the system **100** may protect the program **108a** from a malicious loader. Even if the loader **128** is malicious, the loader **128** cannot pirate the encrypted code **114b-1**, **114b-2**, **114b-3** because it is encrypted. And even if the loader **128** injects additional information into the encrypted code **114b-1**, **114b-2**, **114b-3** or loads the encrypted code **114b-1**, **114b-2**, **114b-3** incorrectly, the authentication server **146** will not verify the hash.

(44) If the authentication server verifies that the processor **136** and the contents of the hardware enclave **132** can be trusted (such as by verifying both the processor signature and the hash), the authentication server **146** may send the decryption key **124** to the computing device **104**. The

processor **136** may use the decryption key **124** to decrypt the encrypted code **114b-1**, **114b-2**, **114b-3**. At that time, the hardware enclave **132** may contain decrypted code **120-1**, **120-2**, **120-3**. The decrypted code **120-1** may be a decrypted version of the encrypted code **114c-1**. The decrypted code **120-2** may be a decrypted version of the encrypted code **114c-2**. The decrypted code **120-3** may be a decrypted version of the encrypted code **114c-3**. The decrypted code **120-1**, **120-2** can be executed even when the decrypted code **120-1**, **120-2** is an arbitrary distance in the memory **130** from the data **116c-1**, **116c-2**. And the decrypted code **120-3** can be executed even when the decrypted code **120-3** is an arbitrary distance in the memory **130** from the data **116c-3**.

(45) Before the processor **136** executes the decrypted code **120-1**, **120-2**, **120-3**, the startup code **134** may cause the processor **136** to mark the hardware enclave **132** as non-readable to all instructions residing inside the hardware enclave **132**. The startup code **134** may include these instructions in order to protect the decrypted code **120-1**, **120-2**, **120-3** from leaking content. Readability and executability may be two separate permissions. Therefore, marking the hardware enclave **132** as non-readable may not prevent the decrypted code **120-1**, **120-2**, **120-3** from being executed. Once the processor **136** marks the hardware enclave **132** as non-readable, the processor **136** may execute the decrypted code **120-1**, **120-2**, **120-3**. The decrypted code **120-1**, **120-2**, **120-3** may perform the same functions as the code included in the files **110a**, **112a**.

(46) FIGS. 2A and 2B illustrate a file **210a** and a modified file **210b** in accordance with the techniques described herein. The file **210a** may be one example of the file **110a** shown in FIG. 1. The modified file **210b** may be one example of the modified file **110b** shown in FIG. 1.

(47) The file **210a** may be part of a COTS software program. The file **210a** may be an unencrypted binary file that includes executable code and data. For example, the file **210a** may include a code section **220a-1**, a code section **220a-2**, a data section **216a-1**, and a data section **216a-2**. The code sections **220a-1**, **220a-2** may include executable code and may be unencrypted. The code sections **220a-1**, **220a-2** may be much smaller in size than the data sections **216a-1**, **216a-2**. The code sections **220a-1**, **220a-2** may include one or more references to the data sections **216a-1**, **216a-2**. For example, the code section **220a-1** may include data reference **242a-1**, data reference **242a-2**, and data reference **242a-3**, and the code section **220a-2** may include data reference **242a-4**. Although the file **210a** includes two code sections and two data sections, in other embodiments a file may contain only one code section and one data section or more than two code sections and more than two data sections. A file may also contain an unequal number of code sections and data sections.

(48) The data references **242a-1**, **242a-2**, **242a-3**, **242a-4** may point to specific data or specific locations in the data sections **216a-1**, **216a-2**. For example, the data reference **242a-1** may point to data **244a-1** in the data section **216a-1**. The data reference **242a-2** may point to data **244a-2**. The data reference **242a-3** may point to data **244a-3** in the data section **216a-2**. The data reference **242a-4** may also point to the data **244a-3**. The data references **242a-1**, **242a-2**, **242a-3**, **242a-4** may be formatted such that the data references **242a-1**, **242a-2**, **242a-3**, **242a-4** will point to the correct data during runtime only if the data sections **216a-1**, **216a-2** are loaded in memory at a fixed and predetermined distance from the code sections **220a-1**, **220a-2**. The data references **242a-1**, **242a-2**, **242a-3**, **242a-4** may include instructions to obtain specific data included in the data section **216a-1** or the data section **216a-2**. The data references **242a-1**, **242a-2**, **242a-3**, **242a-4** may include instructions to modify specific data included in the data section **216a-1** or the data section **216a-2**. Thus, if the data sections **216a-1**, **216a-2** are not loaded at the fixed and predetermined distance from the code sections **220a-1**, **220a-2**, instructions included in the code sections **220a-1**, **220a-2** may obtain or modify the wrong data, and the file **210a** may not function properly.

(49) The modified file **210b** may be a modified version of the file **210a**. A developer of the file **210a** may modify the file **210a** to generate the modified file **210b**. In the alternative, a person or entity other than the developer of the file **210a** (such as a platform) may modify the file **210a** to generate the modified file **210b**. A person or entity other than the developer of the file **210a** may

create the modified file **210b** without access to source code of the file **210a** or debug symbols of the file **210a**.

(50) The modified file **210b** may include encrypted code section **214b-1**, encrypted code section **214b-2**, data section **216b-1**, and data section **216b-2**. The encrypted code sections **214b-1**, **214b-2** may be modified versions of the code sections **220a-1**, **220a-2**. The encrypted code sections **214b-1**, **214b-2** may be encrypted such that a computing device cannot execute the encrypted code sections **214b-1**, **214b-2** without first obtaining a decryption key. The data sections **216b-1**, **216b-2** may be copies of the data sections **216a-1**, **216a-2**. The data sections **216b-1**, **216b-2** may include data **244b-1** (which may be a copy of data **244a-1**), data **244b-2** (which may be a copy of data **244a-2**), and data **244b-3** (which may be a copy of data **244a-3**). The data sections **216b-1**, **216b-2** may be unencrypted plain text.

(51) The encrypted code section **214b-1** may include modified data reference **242b-1**, modified data reference **242b-2**, and modified data reference **242b-3**. The encrypted code section **214b-2** may include modified data reference **242b-4**. The modified data reference **242b-1** may be a modified version of the data reference **242a-1**. The modified data reference **242b-2** may be a modified version of the data reference **242a-2**. The modified data reference **242b-3** may be a modified version of the data reference **242a-3**. The modified data reference **242b-4** may be a modified version of the data reference **242a-4**. The modified data references **242b-1**, **242b-2**, **242b-3**, **242b-4** may be modified such that the modified data references **242b-1**, **242b-2**, **242b-3**, **242b-4** will point to the correct data during runtime even if the distance in memory between the encrypted code sections **214b-1**, **214b-2** and the data sections **216b-1**, **216b-2** is different from a fixed and predetermined distance, which may be defined in the file **210a**. For example, if the data reference **242a-1** points to the data **244a-1** when the data section **216a-1** is placed at a fixed and predetermined distance in memory from the code section **220a-1**, the modified data reference **242b-1** will point to the data **244b-1** when the data section **216b-1** is placed at a distance in memory from the code section **214b-1** that is different from the fixed and predetermined distance. Thus, the modified data references **242b-1**, **242b-2**, **242b-3**, **242b-4** are designed such that there can be an arbitrary distance in memory between the encrypted code sections **214b-1**, **214b-2** and the data sections **216b-1**, **216b-2**.

(52) Modifying the data references **242a-1**, **242a-2**, **242a-3**, **242a-4** may involve identifying and locating the data references **242a-1**, **242a-2**, **242a-3**, **242a-4** within the code sections **220a-1**, **220a-2**. Locating the data references **242a-1**, **242a-2**, **242a-3**, **242a-4** may involve use of relocation information. The file **210a** or a program that includes the file **210a** may include relocation information about all data references within the code sections. The relocation information may be included to support Address Space Layout Randomization (ASLR). ASLR may allow a processor to load a program into an arbitrary place in memory. But ASLR may still require that the distance between code and data be the fixed distance defined in the program. Nevertheless, ASLR may help a platform locate the data references **242a-1**, **242a-2**, **242a-3**, **242a-4**. Thus, the platform may leverage the relocation information to locate data references **242a-1**, **242a-2**, **242a-3**, **242a-4**. After locating the data references **242a-1**, **242a-2**, **242a-3**, **242a-4**, the platform may modify the data references **242a-1**, **242a-2**, **242a-3**, **242a-4** to separate the code sections **220a-1**, **220a-2** from the data sections **216a-1**, **216a-2**.

(53) The modified file **210b** may include a header **218b**. The header **218b** may include information about the modified file **210b**. The header **218b** may be unencrypted. The header **218b** may indicate that the encrypted code sections **214b-1**, **214b-2** have been separated from the data sections **216b-1**, **216b-2** such that the data sections **216b-1**, **216b-2** can be loaded in memory at an arbitrary distance from the code sections **214b-1**, **214b-2**. The header **218b** may indicate that the code sections **214b-1**, **214b-2** should be loaded into a hardware enclave and the data sections **216b-1**, **216b-2** should be loaded in memory outside the hardware enclave. The platform may add the header **218b** to the modified file **210b**.

(54) FIG. 3 illustrates an example method **300** for modifying a file to facilitate protecting a program from piracy using a hardware enclave.

(55) The method **300** may include receiving **302** a file comprising code and data. The file may be part of a program, which may be a COTS software product. The file may be the file **110a**, the file **112a**, or the file **210a**. The program may include additional information other than the file. The file may be an executable file. The file may be an unencrypted binary file. The code may be instructions executable by a processor. The file may be designed such that the data must be stored at a fixed and predetermined distance in memory from the code in order for the file to execute properly. A platform, such as the platform **102**, may receive **302** the file.

(56) The method **300** may include locating **304** data references in the code. The data references may point to specific information or locations in the data. The code may use the data references as part of obtaining or modifying specific information in the data. The data references may be designed such that the data must be stored at a fixed, predetermined distance in memory from the code in order for the data references to point to the correct information in the data during runtime. The platform may locate **304** the data references. The platform may locate **304** the data references without using source code or debug symbols associated with the file. The platform may locate **304** the data references without using information from a developer of the program other than the program.

(57) Locating **304** the data references in the code may include using relocation information about the data references contained in the code, the file, or the program. The relocation information may be included to support ASLR.

(58) The method **300** may include modifying **306** the data references in the code. Modifying **306** the data references may include separating the code from the data. When separated, the data can be placed at an arbitrary distance in memory from the code and the data references in the code will still point to the correct information in the data during runtime. The platform may modify **306** the data references. The platform may modify **306** the data references without accessing source code or debug symbols associated with the file. The platform may modify **306** the data references without accessing information from a developer of the program other than the program.

(59) The method **300** may include encrypting **308** the code. Decrypting the encrypted code may require use of a decryption key. The decryption key may be a private decryption key. The platform may encrypt **308** the code. The platform may not encrypt the data.

(60) The method **300** may include modifying **310** the file to include a separation header. The separation header may indicate that the data in the file can be placed at an arbitrary distance in memory from the code. The separation header may indicate that the code should be loaded in memory inside a hardware enclave and that the data should be loaded in the memory outside the hardware enclave. The platform may modify **310** the file to include the separation header.

(61) The method **300** may include measuring **312** a hash of the encrypted code and startup code. Measuring **312** the hash may include performing a public, standard hash function on the encrypted code and the startup code. The startup code may be designed to be placed in the hardware enclave with the encrypted code. Measuring **312** the hash may include storing the hash as a verified hash value. The platform may measure **312** the hash. The platform may store the hash on the platform or on an authentication server. The hash may be used in authenticating a request for a decryption key to decrypt the encrypted code.

(62) The method **300** may include providing **314** the file for distribution. The file may include the separation header, the modified data references, and the encrypted code. Providing **314** the file for distribution may include providing copies of the file for users to download. The platform may provide **314** the file for distribution. A computing device may download the file from the platform.

(63) FIG. 4 illustrates a potential method **400** for loading an application to facilitate protecting the application from software piracy.

(64) The method **400** may include receiving **402** a request to initiate an application. The application

may include one or more files with executable code. Each of the one or more files may include both code and data. The code and the data in the one or more files may have been separated such that the data from each of the one or more files can be loaded at an arbitrary distance in memory from the code of that file. The code in the one or more files may be encrypted. The data in the one or more files may be unencrypted. An operating system, such as the operating system **126**, may receive **402** the request.

(65) The method **400** may include reading **404** a header from the one or more files. The header may be contained in or associated with the one or more files. The header may indicate that the code and the data in the one or more files are separated. The header may indicate that the code should be loaded in a hardware enclave in a continuous range and the data should be loaded in memory outside the hardware enclave. The operating system may read **404** the header. The operating system may include a loader, such as the loader **128**, designed to read and understand the header.

(66) The method **400** may include loading **406** startup code into a hardware enclave. The startup code may include instructions consumable by a processor. The startup code may include instructions to cause a processor to measure a hash of information stored in the hardware enclave, sign the hash with a signature, provide the hash and the signature to an authentication server, and mark the hardware enclave as non-readable. The operating system may load **406** the startup code into the hardware enclave. The operating system may include the startup code. In the alternative, the operating system may receive the startup code from a platform.

(67) The method **400** may include loading **408** code contained in the one or more files into the hardware enclave. The code may be encrypted. Loading **408** the code may include loading the code in a continuous range in the hardware enclave. The operating system may load **408** the code contained in the one or more files into the hardware enclave. The operating system may use a loader to load **408** the code contained in the one or more files into the hardware enclave.

(68) The method **400** may include loading **410** data contained in the one or more files into memory outside the hardware enclave. The data may not be encrypted. Loading **410** the data may include loading the data in a continuous range in memory outside the hardware enclave. Loading **410** the data may include loading the data at an arbitrary distance in the memory from the code. The operating system may load **410** the data contained in the one or more files into memory outside the hardware enclave. The operating system may use the loader to load **410** the data.

(69) The method **400** may include causing **412** execution of the startup code. The operating system may cause **412** execution of the startup code. The startup code may be unencrypted.

(70) FIG. 5 illustrates an example method for providing a decryption key in accordance with the techniques described herein.

(71) The method **500** may include receiving **502** a request for a decryption key, the request including a hash and a signature. A computing device may make the request and may provide the hash and the signature. The computing device may make the request as part of a remote attestation process. The computing device may include a processor and a hardware enclave. The hardware enclave may include encrypted code and startup code. The processor may have measured the hash of the encrypted code and the startup code. The processor may have signed the hash with the signature using a private certificate. A platform or an authentication server may receive **502** the request.

(72) The method **500** may include authenticating **504** the signature. Authenticating **504** the signature may include determining that the hash was signed using the private certificate. Authenticating **504** the signature may be done using a public certificate. The platform or the authentication server may authenticate **504** the signature. Authenticating **504** the signature may verify that the processor is authentic and can be trusted to enforce the confidentiality and/or integrity protections of the hardware enclave.

(73) The method **500** may include verifying **506** the hash. Verifying **506** the hash may include comparing the hash to a verified hash value. The platform or the authentication server may verify

506 the hash. Verifying **506** the hash may verify that the encrypted code and the startup code stored in the hardware enclave have not been modified by the computing device. Verifying **506** the hash may verify that the encrypted code and the startup code can be trusted. The platform or the authentication server may have determined the verified hash value.

(74) The method **500** may include providing **508** the decryption key. The decryption key may allow the computing device to decrypt the encrypted code. The startup code may include instructions to cause the processor to decrypt the encrypted code. The platform or the authentication server may provide **508** the decryption key. The platform or the authentication server may not provide **508** the decryption key unless the signature is authentic and the hash matches the verified hash value. If the signature or the hash is not verified or validated, the platform or the authentication server may notify the computing device that the platform or the authentication server will not provide the decryption key.

(75) FIG. **6** illustrates an example method **600** for initiating a program modified in accordance with the techniques disclosed herein.

(76) The method **600** may include causing **602** a processor to measure a hash of code in a hardware enclave. Startup code stored in the hardware enclave may include instructions to cause **602** the processor to measure the hash of the code in the hardware enclave. The processor may measure the hash of the code in the hardware enclave using a standard hash function. The code in the hardware enclave may include startup code and encrypted code from a program. The program may include files whose code sections and data sections have been separated. Measuring the hash of the code in the hardware enclave may be part of a remote attestation process to obtain a decryption key. Other methods for attesting to the contents of the hardware enclave may be used other than measuring the hash.

(77) The method **600** may include causing **604** the processor to sign the hash with a signature. The startup code may include instructions to cause **604** the processor to sign the hash with the signature. The processor may sign the hash with the signature using a private certificate. Signing the hash with the signature may be part of the remote attestation process for obtaining the decryption key. Other methods for attesting to the authenticity of the processor may be used other than signing the hash with the signature.

(78) The method **600** may include causing **606** the processor to request a decryption key from an authentication server, the request including the hash and the signature. The startup code may include instructions to cause **606** the processor to request the decryption key. The processor may make the request and provide the hash and the signature to the authentication server over a secure communication channel. The authentication server may be part of a platform. The authentication server may include the decryption key. The authentication server may provide the decryption key if the authentication server verifies the hash and the signature. If the authentication server determines either that the signature is not authentic or that the hash is incorrect, the authentication server may not provide the decryption key. Although the method **600** includes causing **606** the processor to request the decryption key using the hash and the signature, other attestation methods may be used.

(79) The method **600** may include causing **608** the processor to decrypt the code. The startup code may include instructions to cause **608** the processor to use the decryption key to decrypt the code.

(80) The method **600** may include causing **610** the processor to mark the hardware enclave as non-readable. The startup code may include instructions to cause **610** the processor to mark the hardware enclave as non-readable to instructions within the hardware enclave.

(81) FIG. **7** illustrates certain components that may be included within a computer system **700**. One or more computer systems **700** may be used to implement the various devices, components, and systems described herein.

(82) The computer system **700** includes a processor **701**. The processor **701** may be a general purpose single- or multi-chip microprocessor (e.g., an Advanced RISC (Reduced Instruction Set Computer) Machine (ARM)), a special purpose microprocessor (e.g., a digital signal processor

(DSP)), a microcontroller, a programmable gate array, etc. The processor **701** may be referred to as a central processing unit (CPU). Although just a single processor **701** is shown in the computer system **700** of FIG. 7, in an alternative configuration, a combination of processors (e.g., an ARM and DSP) could be used.

(83) The computer system **700** also includes memory **703** in electronic communication with the processor **701**. The memory **703** may be any electronic component capable of storing electronic information. For example, the memory **703** may be embodied as random access memory (RAM), read-only memory (ROM), magnetic disk storage media, optical storage media, flash memory devices in RAM, on-board memory included with the processor, erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM) memory, registers, and so forth, including combinations thereof.

(84) Instructions **705** and data **707** may be stored in the memory **703**. The instructions **705** may be executable by the processor **701** to implement some or all of the functionality disclosed herein. Executing the instructions **705** may involve the use of the data **707** that is stored in the memory **703**. Any of the various examples of modules, components, packages, applications, and operating systems described herein may be implemented, partially or wholly, as instructions **705** stored in memory **703** and executed by the processor **701**. Any of the various examples of data described herein may be among the data **707** that is stored in memory **703** and used during execution of the instructions **705** by the processor **701**.

(85) A computer system **700** may also include one or more communication interfaces **709** for communicating with other electronic devices. The communication interface(s) **709** may be based on wired communication technology, wireless communication technology, or both. Some examples of communication interfaces **709** include a Universal Serial Bus (USB), an Ethernet adapter, a wireless adapter that operates in accordance with an Institute of Electrical and Electronics Engineers (IEEE) 802.11 wireless communication protocol, a Bluetooth® wireless communication adapter, and an infrared (IR) communication port.

(86) A computer system **700** may also include one or more input devices **711** and one or more output devices **713**. Some examples of input devices **711** include a keyboard, mouse, microphone, remote control device, button, joystick, trackball, touchpad, and lightpen. Some examples of output devices **713** include a speaker and a printer. One specific type of output device that is typically included in a computer system **700** is a display device **715**. Display devices **715** used with embodiments disclosed herein may utilize any suitable image projection technology, such as liquid crystal display (LCD), light-emitting diode (LED), gas plasma, electroluminescence, or the like. A display controller **717** may also be provided, for converting data **707** stored in the memory **703** into text, graphics, and/or moving images (as appropriate) shown on the display device **715**.

(87) The various components of the computer system **700** may be coupled together by one or more buses, which may include a power bus, a control signal bus, a status signal bus, a data bus, etc. For the sake of clarity, the various buses are illustrated in FIG. 7 as a bus system **719**.

(88) The techniques disclosed herein can be implemented in hardware, software, firmware, or any combination thereof, unless specifically described as being implemented in a specific manner. Any features described as modules, components, or the like can also be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. If implemented in software, the techniques can be realized at least in part by a non-transitory computer-readable medium having computer-executable instructions stored thereon that, when executed by at least one processor, perform some or all of the steps, operations, actions, or other functionality disclosed herein. The instructions can be organized into routines, programs, objects, components, data structures, etc., which can perform particular tasks and/or implement particular data types, and which can be combined or distributed as desired in various embodiments.

(89) The term “processor” can refer to a general purpose single- or multi-chip microprocessor (e.g., an Advanced RISC (Reduced Instruction Set Computer) Machine (ARM)), a special purpose

microprocessor (e.g., a digital signal processor (DSP)), a microcontroller, a programmable gate array, or the like. A processor can be a central processing unit (CPU). In some embodiments, a combination of processors (e.g., an ARM and DSP) could be used to implement some or all of the techniques disclosed herein.

(90) The term “memory” can refer to any electronic component capable of storing electronic information. For example, memory may be embodied as random access memory (RAM), read-only memory (ROM), magnetic disk storage media, optical storage media, flash memory devices in RAM, on-board memory included with a processor, erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM) memory, registers, and so forth, including combinations thereof.

(91) The steps, operations, and/or actions of the methods described herein may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps, operations, and/or actions is required for proper functioning of the method that is being described, the order and/or use of specific steps, operations, and/or actions may be modified without departing from the scope of the claims.

(92) The term “determining” (and grammatical variants thereof) can encompass a wide variety of actions. For example, “determining” can include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and the like.

(93) The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there can be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. For example, any element or feature described in relation to an embodiment herein may be combinable with any element or feature of any other embodiment described herein, where compatible.

(94) The present disclosure may be embodied in other specific forms without departing from its spirit or characteristics. The described embodiments are to be considered as illustrative and not restrictive. The scope of the disclosure is, therefore, indicated by the appended claims rather than by the foregoing description. Changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A method for protecting software from piracy, the method comprising: receiving a binary file comprising code and data, wherein the code comprises a data reference that is based on a fixed distance in memory between the code and the data; modifying the data reference in the code; and modifying the binary file to indicate that the code of the binary file should be loaded in a hardware enclave and at least a portion of the data of the binary file should be loaded in the memory outside the hardware enclave.
2. The method of claim 1 further comprising: encrypting the code of the binary file but not the data of the binary file; receiving, from a computing device, a request for a decryption key, wherein the computing device includes the hardware enclave; and providing, to the computing device, the decryption key.
3. The method of claim 2, wherein the hardware enclave comprises a defined region of memory of the computing device and the hardware enclave that provides protection from instructions not residing in the hardware enclave.
4. The method of claim 2 further comprising: authenticating, before providing the decryption key, a

processor signature, wherein a processor of the computing device generated the processor signature and the request for the decryption key includes the processor signature.

5. The method of claim 4 further comprising: validating, before providing the decryption key, a hash, wherein the request for the decryption key includes the hash.

6. The method of claim 1, further comprising: identifying the data reference in the code of the binary file without access to source code or debug symbols for the binary file.

7. The method of claim 6, wherein identifying the data reference comprises accessing relocation information about the data reference.

8. A system for facilitating protecting a software program from piracy, comprising: one or more processors; memory in electronic communication with the one or more processors; and instructions stored in the memory, the instructions being executable by the one or more processors to: receive a binary file comprising code and data, wherein the code comprises a data reference that is based on a fixed distance in device memory between the code and the data; modify the data reference in the code; and modify the binary file to indicate that the code of the binary file should be loaded in a hardware enclave and at least a portion of the data of the binary file should be loaded in the device memory outside the hardware enclave.

9. The system of claim 8, wherein modifying the binary file to indicate that the code of the binary file should be loaded in the hardware enclave and at least a portion of the data of the binary file should be loaded in the device memory outside the hardware enclave comprises modifying the binary file to include a separation header.

10. The system of claim 8, the instructions further being executable by the one or more processors to: encrypt the code of the binary file but not the data of the binary file; receive, from a computing device, a request for a decryption key, wherein the computing device includes the hardware enclave; and provide, to the computing device, the decryption key.

11. The system of claim 10, wherein the hardware enclave includes instructions to mark the hardware enclave as non-readable before executing the code.

12. The system of claim 10, the instructions further being executable by the one or more processors to: authenticate, before providing the decryption key, a processor signature, wherein a processor of the computing device generated the processor signature and the request for the decryption key includes the processor signature; and validate, before providing the decryption key, a hash, wherein the request for the decryption key includes the hash and validating the hash comprises comparing the hash to a verified hash value.

13. The system of claim 8, the instructions further being executable by the one or more processors to: identify the data reference in the code of the binary file without access to source code or debug symbols for the binary file.

14. The system of claim 13, wherein identifying the data reference comprises accessing relocation information about the data reference.

15. A non-transitory computer-readable medium comprising instructions that are executable by one or more processors to cause a computing system to: receive a binary file comprising code and data, wherein the code comprises a data reference that is based on a fixed distance in memory between the code and the data; modify the data reference in the code; and modify the binary file to indicate that the code of the binary file should be loaded in a hardware enclave and at least a portion of the data of the binary file should be loaded in the memory outside the hardware enclave.

16. The non-transitory computer-readable medium of claim 15, wherein modifying the binary file to indicate that the code of the binary file should be loaded in the hardware enclave and at least a portion of the data of the binary file should be loaded in the memory outside the hardware enclave comprises modifying the binary file to include a separation header.

17. The non-transitory computer-readable medium of claim 15 further comprising additional instructions that are executable by the one or more processors to cause the computing system to: encrypt the code of the binary file but not the data of the binary file; receive, from a computing

device, a request for a decryption key, wherein the computing device includes the hardware enclave; and provide, to the computing device, the decryption key.

18. The non-transitory computer-readable medium of claim 17, wherein the hardware enclave includes instructions to mark the hardware enclave as non-readable before executing the code.

19. The non-transitory computer-readable medium of claim 18 further comprising additional instructions that are executable by the one or more processors to cause the computing system to: authenticate, before providing the decryption key, a processor signature, wherein a processor of the computing device generated the processor signature and the request for the decryption key includes the processor signature; and validate, before providing the decryption key, a hash, wherein the request for the decryption key includes the hash and validating the hash comprises comparing the hash to a verified hash value.

20. The non-transitory computer-readable medium of claim 15, wherein modifying the data reference does not require access to source code or debug symbols for the binary file.
